
WHEN DO VENTURE CAPITALISTS ADD VALUE?

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EXECUTIVE SUMMARY

Venture capitalists functioning as lead investors and the entrepreneur-CEOs of their portfolio companies responded to questionnaire surveys that asked them to rate the venture capitalists' involvement in the ventures. The perceived effectiveness of the investor's involvement weighted by its perceived importance was used as a proxy for the investor's value to the venture. The survey was administered in the early part of 1988. Eighty

percent of venture capitalists and 85% of entrepreneurs surveyed responded; in all, 51 matched pairs of lead investor-CEO surveys were completed and returned. Over 50 hours of interviews were also conducted to help clarify information derived through the surveys.

CEOs and venture capitalists rated the nature and intensity of their interaction as well as the performance of the venture over a one-year period. CEOs also provided data on strategy, level of innovation, environmental uncertainty, and their level of experience. Regression analyses indicated that a significant portion of variation in the value of venture capitalist involvement was explained by these factors. Specifically, the greater the innovation pursued by the venture, the more frequent the contact between the lead investor and the CEO, the more open the communication, and the less conflict of perspective in the venture capitalist-CEO pair, the greater was the value of the involvement. Neither the stage of the venture nor the CEO's experience had a significant impact on value added. The value of venture capitalists' involvement was also strongly positively correlated with venture performance.

The implications for the venture capitalists include the following: (1) the value of involvement varies with circumstances; (2) the most effective venture capitalists are those who maintain frequent, open communications while minimizing conflict; (3) opportunities exist for adding value in all venture stages; and (4) both experienced and inexperienced CEOs can benefit.

For entrepreneurs, three key findings are: (1) because venture capitalists can add value beyond the money supplied, choosing the right one at the outset is very important; (2) once in, it is important

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to keep communication channels open; and (3) high innovation ventures benefit most from venture capitalist involvement.

The results are important because they provide insight into controllable circumstances that impact the value of venture capitalist involvement in their portfolio companies. Given the general economic conditions now facing entrepreneurs and the degree of cut-throat competition in the venture capital industry, such information may prove extremely useful to both as they plan strategies for the 1990s.

INTRODUCTION

Despite the dramatic growth of venture capital in the U.S. over the last decade, the venture capital industry in the U.S. has been facing rough times, and no relief is currently in sight. The hard facts are that there is less money flowing into the industry than in 1987 and that rates of return have been dwindling since 1983 (Bygrave 1989). More venture capital is currently being raised in Europe than in the U.S., and alternative sources of financing for entrepreneurs have grown dramatically (Timmons and Sapienza 1990).

Though the prescience and sagacity of venture capitalists was being hailed in the mid-1980s, the tune has now changed. Some have argued that the best investment prospects do not generally consider venture capital when searching for financing sources (Amit et al. 1990), while others have charged that venture capitalists at times get in the way of a venture's progress (Rosenstein et al. 1990).

What has happened? In a meeting of the National Venture Capital Association, May 1990, Michael Porter, William Sahlman, and Howard Stevenson of Harvard argued that the influx of new domestic and foreign competitors, the increasing power of institutional investors, the growth of alternative financing sources such as strategic alliances and angels, and the increasing sophistication of entrepreneurs have undermined the structure of the venture capital industry and eroded returns. Because their limited partners have demanded less-risky investments and quicker returns, venture capital (VC) firms responded by making later-stage investments; meanwhile, entrepreneurs continue to seek better investment terms and to demand greater attention from their VC investors.

Timmons and Sapienza (1990) suggest that only those VC firms able to differentiate themselves from the more than 700 other domestic VC firms via value added involvement in their portfolio companies are likely to survive the impending shakeout in the VC industry. There has been little empirical research up to this point on the issue of how venture capitalists add value to the ventures they fund or on how much value is actually added. Such information is clearly of importance to suppliers of capital (limited partners), venture capitalists themselves, users of capital (entrepreneurs), and public policy makers.

EMPIRICAL EVIDENCE

Two general approaches have been taken by researchers on the issue of VC value added: one approach has been to look at early stock returns for VC-backed versus non-VC-backed initial public offerings; the other approach has been to survey entrepreneurs and venture capitalists on their perceptions of value added. We will review findings for each of these approaches.

Cherin and Hergert (1988) examined cumulative and risk-adjusted returns for a 24-month period after initial public offering (IPO) of 71 VC-backed ventures and 59 non-VC-backed ventures in the computer equipment and software industry. Using Compustat data,

they found both sets to be realizing negative returns during this period, with no statistical difference between the two groups, and therefore rejected the notion that venture capitalists provide additional value. As they point out, however, this methodology fails to control for value that may have been added by venture capitalists in the approximately five-year-period before the IPO and which may have already been factored into the initial price.

Brophy and Verga (1988) compared initial stock price and the variability of stock returns for 20 days after IPO of 210 VC-backed and 1,053 non-VC-backed ventures, from 1977 to 1983. They hypothesized that if venture capitalists add value, the companies they back will be less underpriced at the outset than otherwise; further, their early returns would vary less than that of other ventures. This methodology attempts to take into account value that has already been supplied by venture capitalists; they also controlled for the presence of prestigious underwriters in their design. They concluded that: (1) ventures with VC backing outperform others; (2) ventures with VC backing do not gain substantially by adding a prestigious underwriter; and (3) non-VC-backed companies *do* benefit by having a prestigious underwriter.

Stein and Bygrave (1990) examined the returns of 77 VC-backed high-tech companies from the point of the original VC investment through four years after the IPO. They defined the "top 20 high tech venture capitalists" as those firms that held the most seats on the boards of directors of the sample companies (20 VC firms held 84 seats on 51 companies; the remaining 51 VC firms held 80 seats on 26 companies). Based on this split, they found that ventures with venture capitalists from the "top 20" enjoyed higher returns and concluded that venture capitalists from the "top 20" firms do indeed add value. One potential problem with this conclusion is that if these VC firms were among the elite, they may have been able to secure equity at a discounted price because of their reputation and prestige (Perry 1988). This again points up the difficulty of distinguishing the inherent value of the investment and the savvy of those arranging the investment deal versus additional value created through intervention by venture capitalists.

One problem with these studies is that they show only that the VC-backed ventures have (or have not) realized an initial stock price advantage. They do not investigate what actions are more or less valuable to the development of portfolio companies.

While the above studies treat value-added as a black box, other studies attempt to look at the activities of the venture capitalists in the ventures. Rosenstein et al. (1989) received questionnaire responses from 198 VC-backed ventures (26% response rate) in Massachusetts, California, and Texas. They found that the CEOs of these ventures did not perceive that venture capitalists on their boards added more value than did other board members; furthermore, they found no significant correlation between perceptions of value-added and venture performance. In a follow-up study, Rosenstein et al. (1990) conducted telephone surveys with 98 of the 162 companies in their first study. They found that CEOs rated venture capitalists as contributing about the same as other outside board members unless there was a venture capitalist on the board from the "top 20" VC firms (i.e., defined as in Stein and Bygrave 1990); when the "top 20" were represented, venture capitalists were rated as contributing *more* to the venture than outside board members. Rosenstein et al. concluded that venture capitalists add value *only* when there is one from the "top 20" present on the board. This conclusion appears to presume that other outside board members add no value to new ventures.

MacMillan et al. (1989) surveyed 62 venture capitalists about their input into strategic and operational activities relative to that of the top entrepreneurs in the ventures they funded. Using factor and regression analyses, they found that venture capitalists varied in their level of involvement and that high levels of some activities (e.g., negotiating employment terms) were associated with high performance, whereas high levels of other activities (e.g., re-

cruiting top management) were associated with poor performance, depending on the level of involvement of the lead investor. They found no evidence of systematic variation in involvement and concluded that activity levels depended on the personal preferences of the venture capitalists.

While this study begins to address the crucial question of *how* venture capitalists might add or detract value, the cross-sectional nature of the study clouds interpretation of activity-performance relationships. For example, poor venture performance may lead to high venture capitalist involvement in some activities (e.g., management recruiting) rather than resulting from these activities.

Sapienza and Timmons (1989) matched responses of CEOs of VC-backed ventures and lead investors to examine their assessment of the importance of individual roles assumed by venture capitalists. Three role-types emerged: strategic, supportive, and networking. Their results were consistent with those of MacMillan et al. (1989) and Rosenstein et al. (1989) with regard to important key roles (e.g., sounding board, financier, contact, management recruiter) and also identified the importance of non-instrumental roles such as mentor and confidant or friend. They found that venture capitalists' roles were assessed as being more important: (1) in early stage ventures; (2) for entrepreneurs who had less start-up experience; and (3) when venture capitalists held a larger equity position in the venture. Furthermore, the strategic roles were classified by both entrepreneurs and venture capitalists as most important, followed by social/supportive and networking roles, respectively. The study concluded that venture capitalists do add more than money; it did not, however, attempt to link the importance ratings with venture performance.

Gomez-Mejia et al. (1990) conducted 10 qualitative interviews with venture capitalists and 10 with CEOs and concluded that CEOs view the financial and networking activities as valuable, but give mixed reviews to the value of managerial involvement of venture capitalists, with many CEOs viewing managerial intervention as counterproductive.

Inconsistencies in findings can be attributed at least in part to differences in measures used to examine the value of venture capitalist input. Stock-price based measures enjoy the advantage of being objective and being tied to an economic outcome; the causes of the outcomes, however, remain shrouded in mystery. Perception based measures allow insights into the VC process but are potentially biased and do not adequately quantify value added or detracted. In sum, the evidence thus far suggests that in many circumstances venture capitalists do add value; however, key questions remain: When do venture capitalists most add (or detract) value? How much value? This study was designed to help answer the former question.

DIMENSIONS OF INTEREST AND HYPOTHESES

Researchers and practitioners have observed that venture capitalists provide value-added services to ventures (e.g., Kramer 1984; Timmons 1984), but few have examined the circumstances or methods that allow for substantial value creation. The following discussion focuses on context and venture capitalist-CEO (VC-CEO) relations variables expected to impact the value of venture capitalist involvement in the venture.

Context and Value-Added

Stage

The stage of a venture's development has long been seen as a critical variable. For example, venture viability has been observed to vary with stage (Stinchcombe 1965); some have

claimed that venture needs, objectives, and management styles do or should vary with stage (e.g., Churchill and Lewis 1983). Bygrave's (1987) finding that VC firms tend to co-invest more extensively when funding early-stage ventures suggests that venture capitalists view such ventures as riskier than later-stage companies.

These risks result in part from venture newness itself. In the early stages, ventures are less likely to have established relationships with suppliers, distributors, buyers, and other business and professional help. Ventures may function with skeleton staffs, operating with positions unfilled and responsibilities as yet undefined; the CEOs may be untested in the industry or as leaders. For some of these reasons, Timmons and Bygrave (1986) concluded that venture capitalists made their greatest contributions to high-tech ventures in their earliest stages and that the important contributions tended to be managerial rather than strictly financial. Sapienza and Timmons (1989) provided some empirical support for this contention.

In summary, past evidence suggests that VC involvement is more critical to early than to late-stage ventures:

H1: The earlier the stage of the venture, the greater will be the value of venture capitalist involvement.

Innovation

Entrepreneurs seeking to gain market entry or competitive superiority through high innovation relative to competitors face severe information demands and impediments. If they seek to outpace competitors, they must keep abreast of competitor actions, and they must know what the market wants. The greater the number of dimensions upon which they seek to innovate (e.g., technology, marketing, service), the greater these demands become. While Porter (1985) has suggested that innovating along several dimensions of the value chain simultaneously can lead to sustainable superiority, achieving this can be especially problematic for developing businesses that do not have the resources or personnel to commit to the necessary boundary-spanning activities.

Gomez-Mejia et al. (1990) found that all of the entrepreneur-CEOs surveyed agreed that venture capitalists performed valuable boundary-spanning functions. As the number of dimensions in which a venture is attempting to innovate increases, the greater becomes the chance that venture capitalists can provide knowledge and service of value to the venture. Since innovative ventures can make great use of strategic input from outside advisors, such ventures appear to be the ones that can most benefit from lead-investor involvement:

H2: The higher the level of innovation pursued by the venture, the greater will be the value of venture capitalist involvement.

Competitive Strategy

Rosenstein (1988) found that venture capitalists play a major role in strategy formulation and implementation for their portfolio companies. Sapienza and Timmons (1989) also found that both venture capitalists and entrepreneurs attributed great importance to venture capitalists' strategic role in the venture. By functioning as sounding boards and advisors, venture capitalists provide additional perspectives on, and knowledge of, markets and external conditions.

Gupta (1987) argued that implementing a competitive strategy of differentiation re-

quires greater access to external information than does a low-cost strategy. Successful differentiation often depends upon information about the specific needs of suppliers and buyers, potential substitute products, and the attributes of competitor offerings. On the other hand, primary methods for achieving low cost often involve technical process innovation. While the information needed for differentiation is not necessarily any greater than that required for low-cost positioning, venture capitalists are much more likely to be able to contribute information on market conditions than on technical innovation. Thus, the value of venture capitalist input is apt to be greater when the venture is pursuing a differentiation strategy:

H3: The farther the venture is toward the differentiation end of the strategy spectrum, the greater will be the value of venture capitalist involvement.

Environmental Uncertainty

The extent to which an entrepreneur can predict the future states of the environment will depend upon the knowledge he or she has about conditions, the skill in applying such knowledge, and the inherent volatility and instability in the environment. Perceived environmental uncertainty is high when entrepreneurs are unable to achieve a level of confidence in their forecasts. A savvy venture capitalist may be of some help when the primary cause of the uncertainty is from insufficient information or inappropriate application of the information available. When there is greater certainty about demand, supply, prices, and customer preferences, the value of outside input to forecasts is minimal. Therefore, one would expect the value of venture capitalist involvement to be greatest when perceived environmental uncertainty is high:

H4: The greater the perceived environmental uncertainty, the greater will be the value of venture capitalist involvement.

Entrepreneur's Experience

The "quality" of the entrepreneur or entrepreneurial team has often been identified as a key component in venture capitalists' investment decision. An important practical question is, who benefits most by venture capitalist input, experienced or inexperienced entrepreneurs?

Inexperienced entrepreneurs stand to gain a great deal from venture capitalists' wealth of start-up experience; they can gain insights on what to expect at various stages, what to worry about and what not to, and on how they are progressing. This may be extremely valuable information. On the other hand, it may be that what venture capitalists do best is to complement and enhance the skills of experienced entrepreneurs. That is, such entrepreneurs may be more fully able to process and use outside advice, judge which is pertinent and which is not, and to take advantage of the connections and legitimacy afforded through the VC firms.

Therefore, it is unclear whether less experienced or more experienced CEOs are apt to derive greater value through venture capitalist involvement. Thus, no hypothesis can be advanced about the effect of the CEO's experience on the value of venture capitalist involvement. Nonetheless, the experience of the CEO could prove to be an extremely important variable in the potential for the venture capitalist to add value.

VC-CEO Interaction and Value-Added

Frequency of Interaction

A basic condition for venture capitalist involvement to add value beyond the financing provided is that an exchange of information take place in a timely manner. The more frequently interaction occurs, the greater the likelihood that problems and possible solutions are identified within the appropriate time frame.

Information exchange and contact are not costless; they incur opportunity costs and may slow implementation. However, both sides have cause to seek interactions in which the benefits exceed the costs. One would expect that interaction will occur primarily when one or both sides perceive the potential for gain through contact and communication in the VC-CEO dyad. If, in fact, venture capitalist involvement generally adds value to the venture, frequent involvement is likely to be more effective:

H5: The more frequent the interaction in VC-CEO dyads, the greater will be the value of venture capitalist involvement.

All of the above hypotheses are consistent with the information processing perspective (Galbraith 1973), which holds that effective organizations design their information processing structure for efficiency. Thus, when the organization's needs are high or its processing capacity low, more information must be processed or it must be processed with greater rapidity to function properly. In highly complex and unstructured environments such as those faced by entrepreneurial ventures, organic, informal coordinating mechanisms are held to be most efficacious. For example, Gupta (1987) found that in highly volatile environments, more openness in corporate-business unit relations leads to higher effectiveness of the corporate-business unit relations.

The risks, complexities, and uncertainties faced by new high-growth ventures create a highly volatile environment. The information processing perspective suggests that in such circumstances an atmosphere conducive to free exchange of information is apt to be more effective than relying on formal channels. This implies that informal and open relations are appropriate for intra- and interorganizational communications in new ventures. Therefore, one would expect that greater value will be derived from the involvement of the lead investors when communication is open and informal:

H6: The more open the interaction of VC-CEO dyads, the greater will be the value of venture capitalist involvement.

Divergence of Perspectives

The effects of a divergence or conflict of perspectives in the VC-CEO pair on the ability of venture capitalist involvement to add value are difficult to predict. On the one hand, different perspectives may prevent "group-think" in board meetings and thus lead to superior alternative strategies. Given the need for creative and innovative solutions in ventures operating on the tightrope between success and failure, a variety of perspectives can be very useful. On the other hand, high divergence of opinion and approach may hamper implementation. Indecision can be especially dysfunctional in crisis situations when the entrepreneur must act quickly and receive timely financial support from the VC group. Furthermore, severe

divergence of opinion might lead to divisiveness, undermining the flow of information and the spirit of collaboration.

In sum, the level of divergence of VC-CEO perspectives may be an aid or a hindrance to a new venture. However, while it is unclear what relationship conflict will have with the value of venture capitalist involvement, it is potentially important.

Venture Performance and Value-Added

Ultimately, what is discovered about the factors influencing the value of venture capitalist involvement will be especially important if such value has a significant impact on venture performance results. It is not reasonable to expect that all ventures in which the lead investor has contributed significantly experience high performance or to expect that, where they have not, the ventures are failing. However, one would expect, *ceteris paribus*, that value added by lead investors would translate positively into venture performance. On average, then, ventures receiving greater value through venture capitalist involvement should be outperforming those receiving less:

H7: The value of venture capitalist involvement will be positively related to venture performance.

METHOD

The data collected for this study included questionnaire responses from the entrepreneur-CEOs of 51 venture capital-backed ventures in the U.S. matched with the responses from the lead VC investor in each of these portfolio companies. The response rates were 85% from entrepreneurs and 80% from venture capitalists. The ventures, geographically dispersed across the U.S., were primarily high-tech but also included service and low-tech firms. They ranged in sales from zero to \$86 million with a median sales of \$4 million; they were from 1 to 26 years since founding, with a median age of four years. Twenty-nine lead investors from 22 VC firms (primarily on the East Coast) participated; these firms ranged in size from \$7 to \$500 million with a median size of \$49 million.

Measures

Table 1 presents summary statistics on the measures used in this study; the "Appendix" contains detailed descriptions of their construction. Unless otherwise indicated, higher values indicate a greater amount of the variable.

Stage of the Venture (STAGE)

The stage of the venture was measured by the venture capitalist's statement of the current financing stage of the venture. The range was from seed stage to bridge/acquisition, with the mean (3.57) falling about in the middle of the six-point scale.

Level of Innovation Pursued (INNOV)

The level of innovation pursued by the venture was measured as the entrepreneur's assessment on a five-point scale of the venture's level of innovation relative to competitors in each of

TABLE 1 Descriptive Statistics

Variable	Mean	S.D.	Min.	Max.	<i>n</i>
STAGE	3.57	1.65	1.00	6.00	51
INNOV	18.87	3.34	9.00	25.00	51
STRAT	17.04	3.33	8.00	25.00	48
UNCTY	17.76	3.17	12.00	26.40	51
FREQ	19.82	5.20	8.00	36.00	51
OPEN	3.00	0.68	2.00	4.75	51
DIVERG	22.45	4.28	14.00	31.00	51
T-EXP	6.70	5.40	0.00	33.50	47
S-EXP	1.82	0.90	1.00	3.00	50
DIST	112.70	126.22	5.00	600.00	46
VALUE	26.01	7.73	10.73	42.20	49
PERFORMANCE	3.11	0.90	1.30	4.89	51

six different dimensions. The possible range of responses was 6 to 30; the observed range was 9 to 25, with a mean of 18.9.

Competitive Strategy of the Venture (STRAT)

The competitive strategy pursued by each venture was measured as the entrepreneur's assessment of the venture's sources of competitive advantage along five categories, each evaluated on a 1 to 5 scale. These were combined to create a low cost-differentiation spectrum with a possible range of extreme low cost, 5, to extreme differentiation, 25; the observed range was 8 to 25, with a mean of 17.

Environmental Uncertainty (UNCTY)

Entrepreneurs' assessment of the unpredictability of seven aspects of the competitive environment (on a 1–5 scale) were summed to measure environmental uncertainty with a potential range of 7 to 35; the observed range was 12 to 26.4, with a mean of 17.8.

Frequency of Interaction (FREQ)

The frequency of contact in VC-entrepreneur (VC-CEO) dyads was measured as the frequency of contact face-to-face, by telephone, and in writing as assessed by both members of the dyad. The numbers in the table are difficult to interpret in the current form, but they indicate a range of contact from about four contacts per year to contact two to three times per week; the mean was contact about three times per month.

Openness in Interaction (OPEN)

Openness in interaction was measured as the average of four responses (two by each member of the dyad) on a 1 to 5 scale indicating the level of friendliness and social interaction outside

of work-related activities for the pair. The possible range was 1 to 5; the observed range was from 2 to 4.75, with a mean of 3.

Value of VC Involvement (VALUE)

The value of the venture capitalist's involvement in the venture was a weighted measure of both members' assessment of the VC's effectiveness in carrying out eight different roles in the venture. The possible range of scores was 1 to 50; the observed range was 10.7 to 42.2, with a mean of 26.

Venture Performance (PERFORMANCE)

Both members of the dyad rated venture performance using a multi-criterion measure (including 15 financial and nonfinancial dimensions); this measure was combined with each respondents' assessment of the overall performance of the venture. The possible range of scores was 1 to 5; the observed range was 1.3 to 4.9, with a mean of 3.1.

Other Variables of Interest

To allow for the possibility that divergence of perspective (DIVERG) in VC-CEO pairs, the CEO's task experience (T-EXP), or his/her start-up experience (S-EXP) account at least partially for differences in the value of venture capitalist involvement, variables measuring these constructs were created. The rationale for inclusion of these variables is explained above in the "Hypothesis" section. Geographic distance (DIST) is also included as a control variable because it may represent an impediment to the timely exchange of information or to the establishment of trust or familiarity necessary to create the proper channels of communication. The method of constructing these measures is described in the "Appendix."

Limitations of Measures in this Study

Because the measures in this study rely on the perceptions of the CEOs and the venture capitalists, caution must be exercised in their application and interpretation. Clearly, for many of the variables of interest (e.g., the nature of interaction, its frequency, and the performance of the portfolio company), no data are publicly available. Nonetheless, bias and inaccuracy are potential threats whenever perceptual measures are used. Accordingly, this study utilized assessments of both the CEO and the venture capitalist to measure the performance of the venture, the value of the involvement, and the interaction dimensions; the fact that these assessments were highly consistent with one another lends credibility to their validity (see the "Appendix" for details). Furthermore, the suggestion of John and Reve (1982) that measures be obtained from a knowledgeable "key spokesman" is followed by using the CEOs' assessment of innovation, strategy, and environmental uncertainty. That their assessments were relative to industry norms helped to control for variations by industry. Still, the possibility that the measures suffer from bias or inaccuracy must be taken into account in interpreting results.

RESULTS

Table 2 presents zero-order correlations among the independent variables, including the four additional control variables.

TABLE 2 Zero-Order Correlations: Independent Variables

Correlations:	1	2	3	4	5	6	7	8	9
STAGE									
INNOV	-0.08								
STRAT	0.05	-0.19							
UNCTY	0.19	-0.03	-0.11						
FREQ	-0.48 ^a	0.15	-0.17	-0.01					
OPEN	-0.11	0.12	-0.03	-0.01	0.36 ^b				
DIVERG	-0.20	0.27 ^c	-0.07	0.00	0.09	-0.28 ^b			
T-EXP	0.26 ^c	0.08	0.06	0.28 ^c	-0.40 ^a	-0.03	0.00		
S-EXP	-0.30 ^b	0.26 ^c	0.09	-0.08	-0.06	-0.17	0.28 ^c	-0.10	
DIST	0.35 ^b	-0.08	0.08	0.06	-0.34 ^b	-0.28 ^c	0.17	0.28 ^c	-0.27 ^c

^a*p* < 0.01.^b*p* < 0.05.^c*p* < 0.10.

These are two-tailed values.

As can be observed from Table 2, the independent variables are not highly intercorrelated as a rule; this suggests that multi-collinearity will not pose severe impediments to the interpretation of the regression analyses. Some interesting observations can be derived, nonetheless, from an examination of Table 2. First, the fact that DIST and STAGE are positively related indicates a tendency for early-stage investments to be made primarily in ventures located near the lead investor. Second, the strong negative correlation between STAGE and FREQ confirms the many anecdotal reports that venture capitalists tend to be most heavily involved in the early stages of the venture's development. For the most part, the correlations are what one might well expect: FREQ and OPEN are positively related; FREQ is lower when the entrepreneur's T-EXP is high; DIVERG in VC-E pairs is negatively related to OPEN; entrepreneurs with greater S-EXP pursue higher INNOV.

Table 3 includes the results of two regression equations (one containing only the six hypothesized independent variables, and the other also containing the four control variables) testing Hypotheses 1 to 6; Table 4 presents the zero-order correlation between VALUE and PERFORMANCE used to examine Hypothesis 7. As can be seen, only one of the control variables have an impact on the value of venture capitalist involvement, less DIVERG is associated with a higher value of venture capitalist involvement (VALUE). Table 3 indicates that the addition of the control variables to the equation does not change the basic relationships

TABLE 3 Tests of Hypotheses^a

(H1) STAGE	(H2) INNOV	(H3) STRAT	(H4) UNCTY	(H5) FREQ	(H6) OPEN	DIVERG	T-EXP	S-EXP	DIST	R ²	F
-0.15	0.16 ^b	0.06	0.05	0.40 ^a	0.29 ^c					0.46	6.3 ^c
-0.17	0.22 ^d	0.06	0.08	0.36 ^c	0.25 ^c	-0.19 ^b	-0.15	0.05	0.01	0.51	4.2 ^e

^aExplaining VC value-added: results of regression analyses. Dependent variable VALUE (role effectiveness weighted by role importance).^b*p* < 0.10.^c*p* < 0.05.^d*p* < 0.01.^e*p* < 0.001.

For T-tests, these are one-tailed values.

TABLE 4 Test of Hypothesis 7^a

	Performance
Value	0.44 ^b

^aRelationship between VC value-added and venture performance, zero-order correlation.

^b $p < 0.001$, one-tail test.

between the predictors and VALUE; it can also be observed that neither T-EXP nor S-EXP has a significant impact on VALUE, though a lower level of T-EXP is moderately associated with greater VALUE of venture capitalist involvement.

Test of Hypothesis 1

Table 3 indicates that the predicted relationship between stage and the value of involvement is *not supported*. While the beta coefficients for STAGE are negative (as predicted), they are not statistically significant.¹

Test of Hypothesis 2

Hypothesis 2 predicted that the value of lead investor involvement would be greater in firms pursuing greater innovation relative to competitors. This hypothesis receives at least *moderate support* in both equations: betas for the effect of INNOV on VALUE = 0.16 ($p < 0.10$), and 0.22 ($p < 0.05$), respectively.

Test of Hypothesis 3

Hypothesis 3 predicted that ventures pursuing strategies toward the differentiation end of the low cost-differentiation spectrum would derive greater value out of venture capitalist involvement. This hypothesis is *not supported*; the betas for STRAT are in the expected direction, but are quite low.

Test of Hypothesis 4

Hypothesis 4 said that as uncertainty increased, the value of venture capitalist involvement would increase. This hypothesis is *not supported*; the betas for UNCTY are in the expected direction, but are quite low.

Test of Hypothesis 5

Hypothesis 5 predicted that the greater the frequency of interaction in the VC-CEO pair, the greater would be the value of the venture capitalist's involvement in the venture. This hypothesis is *supported* by the results: the betas for the effects of FREQ on VALUE = 0.40 ($p < 0.01$) and 0.36 ($p < 0.05$).

¹The lack of a significant association between stage and value should not be construed as definitive evidence refuting the popular notion that venture capitalists' most important contributions are made in earlier stages of the venture's development. A separate analysis indicates that the zero-order correlation between STAGE and VALUE is -0.40 ($p = 0.003$). However, this association does not show up in the regression analyses that control for the frequency of interaction in examining association (recall that the correlation of frequency and stage is -0.48 ; see Table 2).

Test of Hypothesis 6

Hypothesis 6 held that the more open or informal the relations in the dyad, the greater would be the value of venture capitalist involvement. The results, positive and significant, *support* this hypothesis; the betas for the effects of OPEN on VALUE = 0.29 ($p < 0.05$) and 0.25 ($p < 0.05$).

Test of Hypothesis 7

Table 4 shows the zero-order correlation testing the notion that the value of venture capitalist involvement is positively associated with venture performance. The results *support* this view. As predicted, there is a positive relationship ($r = 0.44$, $p < 0.001$) between VALUE and PERFORMANCE; furthermore, this association is quite strong.

CONCLUSIONS AND IMPLICATIONS

This article set out to extend our understanding of the VC process by examining the circumstances under which venture capitalist involvement in portfolio companies contributes more versus less value in the eyes of two of the key stakeholders in the venture: its CEO and the lead venture capital investor. The most important findings are that the level of innovation pursued by the venture and the intensity and style of interaction have a significant impact on the value of venture capitalist involvement.

Conclusions

The connection identified here between high innovation and VC value-added is consistent with Timmons and Bygrave's (1986) observation that venture capitalists make significant contributions beyond the money provided to highly innovative technical ventures. In fact, the positive links between VC value-added and innovation, intensity, and openness of involvement fit perfectly with Perry's (1988) notion that the quality of the VC-E relationship is particularly crucial when the entrepreneur is what Perry calls an "innovator" type: "[Innovators] want a venture capital firm to provide them with money and counsel but, more importantly, they seek a long term partner" (p. 209). Perry also suggested that shared vision, interdependence, and collaboration are preferred conditions for relationships involving innovators. The finding that divergence of perspectives is associated with lower value-added suggests that shared vision may be important across the entire spectrum of entrepreneurial types, not just innovators.

What should be made of the strong positive correlation between value-added and venture performance? One possible explanation is that venture performance may influence perceptions. This possibility cannot be dismissed altogether in a cross-sectional study such as this. However, including ratings from both sides of the VC-E dyad on both measures provides a check on self-rating bias (i.e., we avoid the problem that entrepreneurs or venture capitalists might attribute high performance solely to their own efforts and low performance solely to the failings of the other member of the dyad). The alternative explanation is that, in fact, when venture capitalists are perceived as adding value, they do; and, this value is translated into superior venture performance.

Stein and Bygrave (1990) and Rosenstein et al. (1990) introduced the idea that entrepreneurs may derive value from VC presence only when a venture capitalist from an elite

set of firms ("the top 20") is present on the board of directors. They also suggested that the discrepancies in findings about value-added might be attributable to variations in samples: i.e., those studies that find significant value-added might be composed of VC firms from this elite set. In order to examine this possibility, we examined the available information. Only 4 of the 50 ventures in the current study had lead investors from the list of 14 of "the top 20," which was presented by Stein and Bygrave (1990). The fact that the median size of the VC firms in this study (\$49 million) is well above the median for domestic VC firms in 1987 leaves open the possibility that the VC firms in this sample are "elite" (for instance, Bygrave (1987) presented information that suggested strong competitive advantages for the largest 61 VC firms in the U.S.). However, further investigation of our data revealed that the size of VC firms was not associated with the venture performance or with ratings of the value of venture capitalist involvement. Thus, while it does not refute the idea, this study does not support the notion that only an elite subset of VC firms add value to their investments.

This study has added to our understanding of the contingencies influencing venture capitalist value-added. Here are our conclusions based on this and previous studies: (1) the provision of money alone appears to play a necessary but far-from-sufficient condition to promote economic growth and resilience; evidence is mounting that venture capitalists do add value; (2) some "elite" VC firms *may be* better at this process than others; (3) venture capitalists provide assistance particularly useful for highly innovative ventures; and (4) the nature and style of VC-CEO interactions have a significant impact on the value of venture capitalist involvement; specifically, more involved, open relations appear superior.

Implications

For entrepreneurs of high potential ventures, the growing set of alternative sources of financing provide an attractive array of opportunities. The results here suggest that choosing VC can be particularly rewarding if the entrepreneur is planning to pursue high levels of innovation as a means of gaining competitive advantage. Furthermore, the results here suggest that entrepreneurs should be concerned not only with the money they receive as part of the deal structure but also the amount of assistance that the lead investor and his or her firm is planning to provide. Although venture capitalists may fall along a spectrum of styles from "laissez faire" to "close trackers" (as MacMillan et al. (1989) categorize them), entrepreneurs tend to credit those who are more deeply involved with the venture as being more valuable. Entrepreneurs should carefully assess how well they get along with the individual to be the lead investor and the reputation of the VC firms before committing. The findings also suggest that venture capitalists can add value regardless of the entrepreneur's level of prior experience; thus, entrepreneurs who presume that their experience precludes the possibility of gain through venture capitalist involvement may be doing themselves a disservice. Finally, the evidence suggests that CEOs who work toward an open, partnering atmosphere derive the greatest value from the venture capitalist's involvement.

For venture capitalists, this study suggests that, in general, more involvement is better than less. If this is so, firms that try to reduce risk though increasing diversification may be at a competitive disadvantage: providing value-added services across a wide range of investments and investment types is very difficult. VC firms seeking to differentiate themselves from the burgeoning set of domestic and international firms may do so most productively by focusing and seeking to provide value-added assistance to their portfolio companies. An earlier study found that conflict of perspectives between venture capitalists and entrepreneurs tended to be greatest in ventures seeking high technological innovation (Sapienza and Gupta

1989). The joint implication of that finding and this study is that venture capitalists who are particularly careful about managing the style and nature of their interactions with innovative entrepreneurs are apt to be most effective and valuable.

From a research perspective, several key challenges remain. Specifically, what other contingencies influence value-added? How much value is really added (i.e., controlling for the extra cost of venture capital)? How should value be measured? How long can it be sustained? On what does sustainability depend? What creative solutions for deal structuring exist? and, What public policy initiatives can foster the type of value-added investor involvement necessary for sustained value creation?

VC firms face severe competitive challenges as they enter the 1990s. Perhaps greatest among these will be the challenge of reversing the downward rates of return spiral that have been plaguing them since the early 1980s. At the same time, while new sources of financing developed in the late 1980s for entrepreneurs, the recent threats of recession do not bode well for the availability of capital in the near future. Both sides, important cogs in the development and creation of economic renewal, stand to benefit from further research that yields a deeper understanding of the forces that make their interaction more rather than less productive.

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APPENDIX

Stage

Data to measure the stage of the venture were obtained from venture capitalists. They indicated which of six definitions in Pratt's *Guide to Venture Capital Sources* described the venture's stage: 1 = "seed" financing; 2 = "start-up"; 3 = "first stage"; 4 = "re-start-up"; 5 = "expansion"; 6 = "bridge/acquisition." Measures collected from the CEOs were used for validation. As expected, the measure correlates positively with the following information about the venture provided by CEOs: the number of employees, $r = 0.59$, $p < 0.001$; the time since founding, $r = 0.41$, $p < 0.01$; the time since initial venture capital investment, $r = 0.40$, $p < 0.01$; and sales revenue level, $r = 0.27$, $p < 0.05$.

Innovation

Data on innovation pursued in the venture were obtained from the CEOs. They were asked to indicate, on a five point scale ranging from 1 = "to no degree" to 5 = "to a great degree," the extent to which they were seeking innovation relative to competition in processing technology, product design, management, packaging, distribution, and marketing/sales. These items were based on Porter's (1985) explication of the value chain. The summed score of these six items provided a measure of the level of innovation sought by each venture, with higher totals indicating a greater degree of innovation pursuit for the venture ($\alpha = .54$). Because these dimensions are somewhat independent, a high alpha was not anticipated.

Competitive Strategy

Data on the venture's competitive strategy were obtained from the CEOs. Using a five-point scale ranging from 1 = "significantly lower" to 5 = "significantly higher," CEOs were asked to indicate their venture's position relative to competitors in terms of cost per unit, selling price (or intended selling price), product quality, R&D spending, and marketing

spending. The sum of responses provided a measure of each venture's competitive strategy, with high values indicating differentiation positions and low values indicating low cost positions ($\alpha = 0.63$). This multi-item scale was similar to the one devised by Gupta (1987) to examine the impact of competitive strategy on intra-organizational relations.

Environmental Uncertainty

Data on the uncertainty of the environment surrounding the venture were obtained from the CEO. They were asked to indicate their perceptions of how well they could predict facets of the environment on a five-point scale ranging from 1 = "highly predictable" to 5 = "highly unpredictable." They were asked to assess the predictability of input factors (raw material prices and raw material availability), throughput factors (process technology development and product attributes/design), and output factors (market growth rate, customer needs, and competitors' actions). Thus, a seven-item scale was created; the sum of the items provided a measure of perceived environmental uncertainty, with high values indicating relatively more uncertainty and low values indicating relatively less uncertainty ($\alpha = 0.57$). Because these dimensions of the environment are relatively independent of one another, a high alpha was not anticipated.

Frequency of Interaction

Data to measure the frequency of interaction were obtained from *both* respondents. A seven-point scale (1 = "every day," 2 = "twice a week," 3 = "once a week," 4 = "twice a month," 5 = "once a month," 6 = "once a quarter," and 7 = "less often") was used to indicate the frequency of their contact face-to-face, by telephone, and in writing. A multi-item, multi-rater measure was obtained for frequency of interaction by combining the responses. To check for interrater reliability, the correlations between the responses of the CEO and the venture capitalist for each item was examined; for all three items (i.e., face-to-face, telephone, and written contact), the responses were positively correlated and were significant (at $p < 0.001$, $p < 0.001$, and $p < 0.01$, respectively). Because the responses were assigned lower numbers for greater frequency, the items were reverse scored; when summed, a measure of frequency of interaction was obtained; high values indicate greater frequency and low values indicate less frequency of interaction ($\alpha = 0.86$).

Openness of Relations

Data to measure the openness of VC-CEO relations were obtained from *both* respondents. Using a five-point scale ranging from 1 = "strongly agree" to 5 = "strongly disagree," both members of the VC-CEO dyad were asked to indicate whether or not they agreed that: (1) "this [investor or CEO] and I are very friendly," and (2) "aside from work-related functions, this [investor or CEO] and I do not have frequent social interaction" (reverse scored). A multi-item, multi-rater measure was obtained for openness of interaction by combining the responses. To check for interrater reliability, the correlations between the responses of the CEO and the venture capitalist each item was examined; for each of the two items, the responses were positively correlated and were significant (at $p < 0.002$, and $p < 0.02$, respectively). The summed responses of the four items provided a measure of the relative openness of interaction in the VC-CEO dyad with higher values indicating more open, informal relations ($\alpha = 0.61$).

Value of Venture Capitalist Involvement

Data on the venture capitalist's involvement in the venture were obtained from *both* respondents. They rated on a 1 to 5 scale the importance (1 = none at all to 5 = of great importance) of the lead investor's involvement in *eight key roles* (financier, source of professional contacts, source of industry contacts, business consultant, management recruiter, sounding board, coach/mentor, and friend/confidant) over the last one-year period; for each of these roles, both also rated on a 1 to 10 scale (1 = "very low effectiveness" to 10 = "very high effectiveness") the effectiveness of the venture capitalist in carrying out these roles. The product of the importance and effectiveness ratings of the two respondents were averaged for each role; then, the weighted average across the eight roles provided a scale of the "value" of the venture capitalist's involvement, with higher values indicating more value supplied. The possible range was thus 1 to 50. T-tests indicate no differences between raters, and correlations indicate high inter-rater reliability ($r = 0.67$, $p < 0.001$).

Venture Performance

Data on venture performance were obtained from both respondents. It was measured by (1) a multi-criterion scale and (2) an overall rating of performance. The multi-criterion scale was created by asking both respondents to rate the importance of 15 financial and non-financial dimensions of performance by distributing 100 points across these dimensions according to their estimated relative importance. Then, responding on a 1 to 5 scale, they both provided an assessment of his/her satisfaction with the venture's performance along that dimension (1 = "not at all satisfied," to 5 = "extremely satisfied"). A weighted average measure of performance was obtained for each respondent by multiplying the importance scores by the satisfaction scores and summing. The possible range was 100 to 500; converted to 1 to 5 by dividing by 100. T-tests indicate that the mean response provided by CEOs (2.99) is not significantly different from that provided by venture capitalists (3.25); thus, self-rating bias on the part of CEOs does not appear to be a problem on this measure. To check for interrater reliability, the correlations between the responses were examined; the responses were positively and significantly correlated ($r = 0.44$, $p < 0.001$). Both were also asked to rate the venture's "overall performance" on the same 1 to 5 scale used previously. The mean ratings provided by CEOs (3.12) and those provided by VCs (3.25) were not significantly different. To check for interrater reliability, the correlations between the responses of the CEO and the VC were examined; the responses were positively and significantly correlated ($r = 0.65$, $p < 0.001$). Finally, the four items (the weighted average multi-criterion response from both respondents plus the overall rating from both, were combined and averaged to produce a multi-item, multi-rater measure of venture performance on a five-point scale with higher values indicating relatively superior venture performance ($\alpha = 0.87$).

CEO's Task Environment Experience

CEO's responses were used to measure their task environment experience. The task environment was conceptualized as having three dimensions: the market environment, the technological environment, and the managerial environment. Market experience was measured as the number of years experience the CEO had with the current industry, with current suppliers, and with current customers. Technology experience was measured as the number

of years experience they had with the current process and with the current product technology. Their team experience was measured as the number of years they had worked on the current venture's management team and the number of years working with the current members of the venture's management team, whether or not such work occurred on this venture. The average of the three dimensions were summed to measure the CEO's overall task environment experience ($\alpha = 0.86$).

CEO's New Venture Experience

Data on the extent of CEOs' new venture experience were obtained from the CEOs. CEOs were asked to indicate their new venture experience as: (1) no prior new venture experience; (2) worked in start-ups before but had never been a founder or co-founder of one; or (3) founded a venture prior to the current venture. These responses provided an ordinal scale measuring their new venture experience, with higher values indicating more and lower values less new venture experience.

Divergence of Perspective

Data to measure the divergence of views in VC-CEO interactions were obtained from *both* respondents who used a five-point scale ranging from 1 = "strongly agree" to 5 = "strongly disagree." Respondents were asked to indicate the extent they agreed that: (1) "this [investor or CEO] and I bring different perspectives to most work-related issues," (2) "striving for complete agreement between the two of us would slow down decision making considerably," (3) "this [investor or CEO] and I think alike on most issues" (reverse scored) and (4) "I am extremely satisfied with how well this [investor or CEO] and I are able to resolve conflicts" (reverse scored). Correlations between the responses in the dyad were examined as an indication of interrater reliability; all but one of the correlations were significant at $p < 0.001$. The eight responses were summed to create a measure of divergence of perspectives in the dyad (the possible range was 8–40), with higher values indicating relatively more divergence of perspectives in VC-CEO dyad ($\alpha = 0.63$).

Geographic Distance

Data on the distance between the investor and the venture were obtained from investors. They estimated the time it took them to travel to the portfolio company (expressed as minutes in Table 1).